

ENTRY 026 Baseline-Subtracted Residual Analysis — August 13, 2026 — Status: COMPLETE

Entry 026 — Most of the Forgiveness Ratio Was the Baseline All Along

Subtracting Entry 025's Closed Form From Every Recorded Decay Curve Leaves Almost Nothing Behind

1. What Was Done

Entry 025 derived a closed-form baseline for Bell-circuit forgiveness — $\text{success} = 0.5 + 0.5(1-p)^n$ — from pure depolarizing error alone, no decoherence, no chip identity. Entries 017–021's fitted forgiveness ratios were measured against the full hardware noise model, which includes that baseline plus whatever real chip physics sits on top of it. This entry subtracts one from the other, across every Bell decay curve on record — ten edges spanning Cairo, Sherbrooke, and Kyiv — to see what's actually left.

For each curve: fit k_{measured} the same log-linear way the original entries did; fit k_{baseline} the identical way on the closed-form formula evaluated at the same gate counts, so the sampling matches exactly even though the baseline isn't a pure exponential; then $\text{residual} = (k_{\text{measured}}/p) - (k_{\text{baseline}}/p)$.

2. Result

edge	raw err	D	ratio (orig)	ratio (baseline)	residual
cairo (24,25)	0.600%	4.32	0.5304	0.4704	+0.060
cairo (19,22)	3.132%	0.31	0.3337	0.3502	-0.017
sherbrooke (60,61)	0.347%	2.34	0.6665	0.4829	+0.184
sherbrooke (66,73)	3.067%	9.78	0.4200	0.3531	+0.067
kyiv (119,120)	0.311%	1.12	0.4345	0.4846	-0.050
kyiv (28,29)	0.601%	0.14	1.5904	0.4703	+1.120
kyiv (77,78)	0.986%	3.73	0.5353	0.4513	+0.084
kyiv (45,46)	1.618%	1.65	0.4308	0.4204	+0.010
kyiv (66,67)	2.618%	4.03	0.3937	0.3732	+0.021
kyiv (23,24)	4.323%	5.01	0.2927	0.3009	-0.008

On the seven D-screened usable edges ($D \geq 1.5$), residuals range from -0.008 to $+0.184$ with inconsistent sign, mean $+0.060$. No power law could be fitted in log-log space — some values are negative. Seven points scattered near zero across a 0.35% – 4.32% error span, without a consistent trend, is what noise looks like.

3. An Honest Complication

The D-screen prediction — that decoherence-contaminated edges ($D < 1.5$) should show large residuals, since the baseline has no decoherence term — only partly held. kyiv(28,29), the single worst case ($D = 0.14$, $T_2 = 12.7 \mu\text{s}$), showed a residual of **+1.120**, consistent with severe unmodeled decoherence. But cairo(19,22) ($D = 0.31$) and kyiv(119,120) ($D = 1.12$), both also decoherence-affected, showed small *negative* residuals instead.

The cause is not physics — it's the metric. Residual divides by raw calibrated error p . kyiv(28,29)'s $p = 0.601\%$ is roughly $5\times$ smaller than cairo(19,22)'s $p = 3.132\%$. The same absolute amount of unmodeled decoherence, divided by a five-times-smaller denominator, produces a residual an order of magnitude larger — not because the underlying physics differs by that much, but because the ratio metric is numerically unstable exactly where p is small. This is the same small-denominator fragility that motivated the D-screen in Entry 021, resurfacing in the residual analysis regardless of whether decoherence contamination is present.

Entry 026 — Conclusion

Subtracting the Entry 025 closed-form baseline from every recorded Bell-circuit decay curve leaves, on the seven properly screened edges, no discernible power-law structure — residuals scattered near zero with inconsistent sign. The most defensible reading is that Entries 017–021's empirically fitted forgiveness ratio was mostly re-deriving the fixed geometric baseline, not discovering new hardware physics on top of it. The one edge with a genuinely large residual, kyiv(28,29), is explained by a numerical artifact of the ratio metric at small p , not by an inconsistency in the underlying physics — its raw decoherence severity ($D = 0.14$) is comparable to, not dramatically worse than, cairo(19,22)'s ($D = 0.31$), yet the ratio-based residual differs by more than an order of magnitude. Ratio metrics should not be trusted at low raw error, independent of D .

4. What This Means for QuantumBridge

`variable_forgiveness_ratio()` should be retired, not re-fitted. With no clean residual signal to swap in, the empirically fit power law (Entries 017–021) was mostly measuring the closed-form baseline under a different name. The correct replacement is the derived formula itself — deterministic, requires no calibration data to fit, and is already verified to within shot noise.

The real physical signal — decoherence — is already handled correctly elsewhere: v4's T2-based dephasing term (Entries 023–024), which reads coherence times directly rather than inferring them indirectly through an unstable ratio. Asking a forgiveness-ratio fit to implicitly capture decoherence was always going to be fragile at low error magnitudes; this entry shows exactly why.

5. Next Step — Entry 027

Replace `variable_forgiveness_ratio()` in v4 with the closed-form baseline directly, remove the empirical power-law fit and its two hardcoded anchor points entirely, and re-validate against the Entry 022 Kyiv reference circuits to confirm the simplification doesn't regress Entry 024's routing fix. If it holds, the six-edge decay-curve protocol itself may no longer be necessary for future chips — only the D-screen and T2/T1 calibration export would be, considerably cutting the calibration cost the project has been carrying since Entry 013.

Research Log — Index Addendum (Entry 026)

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Entries 017-021's forgiveness-ratio power law is superseded: not wrong as a measurement, but shown to be mostly re-deriving the Entry 025 geometric baseline rather than capturing independent chip physics.